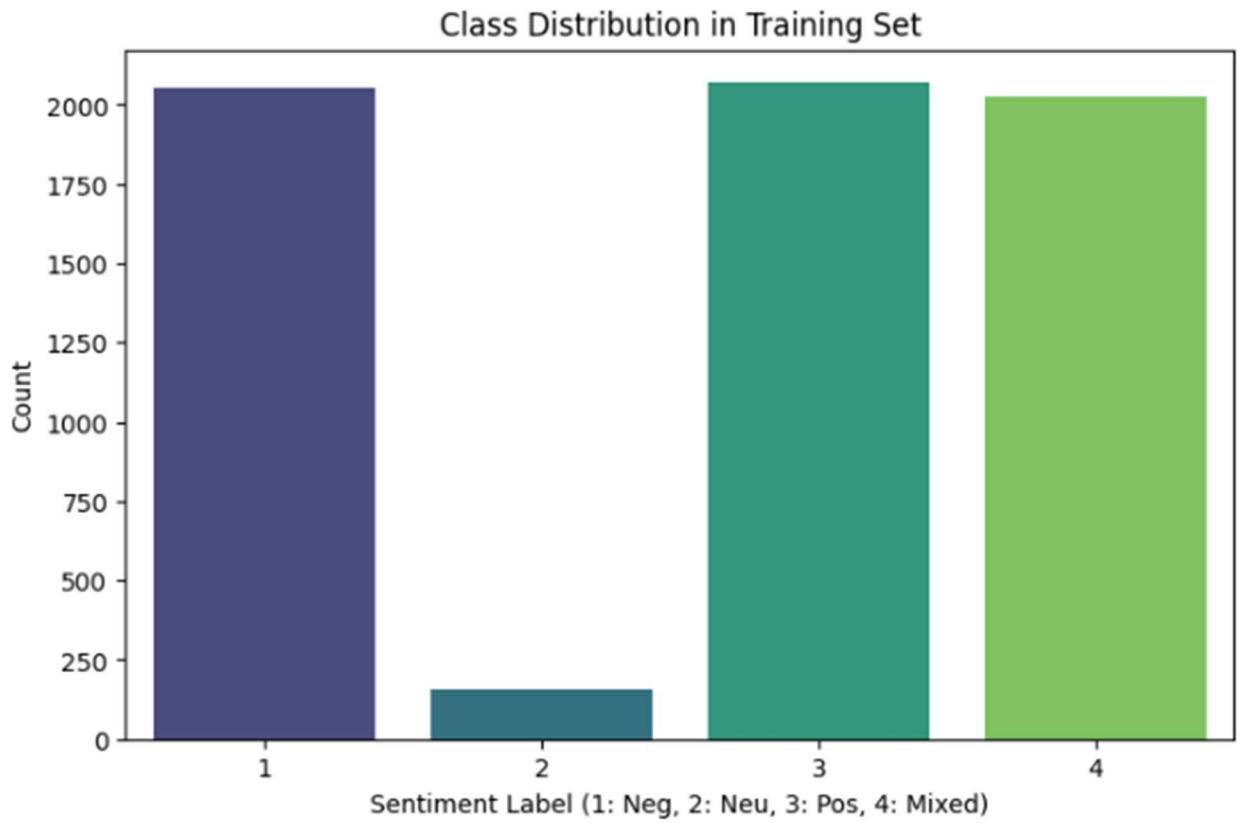
Lab Exercise 3: Word2Vec and Multiclass Classification

1. Exploratory Data Analysis (EDA)

The dataset consists of product reviews labeled into four sentiments: Negative (1), Neutral (2), Positive (3), and Mixed (4).

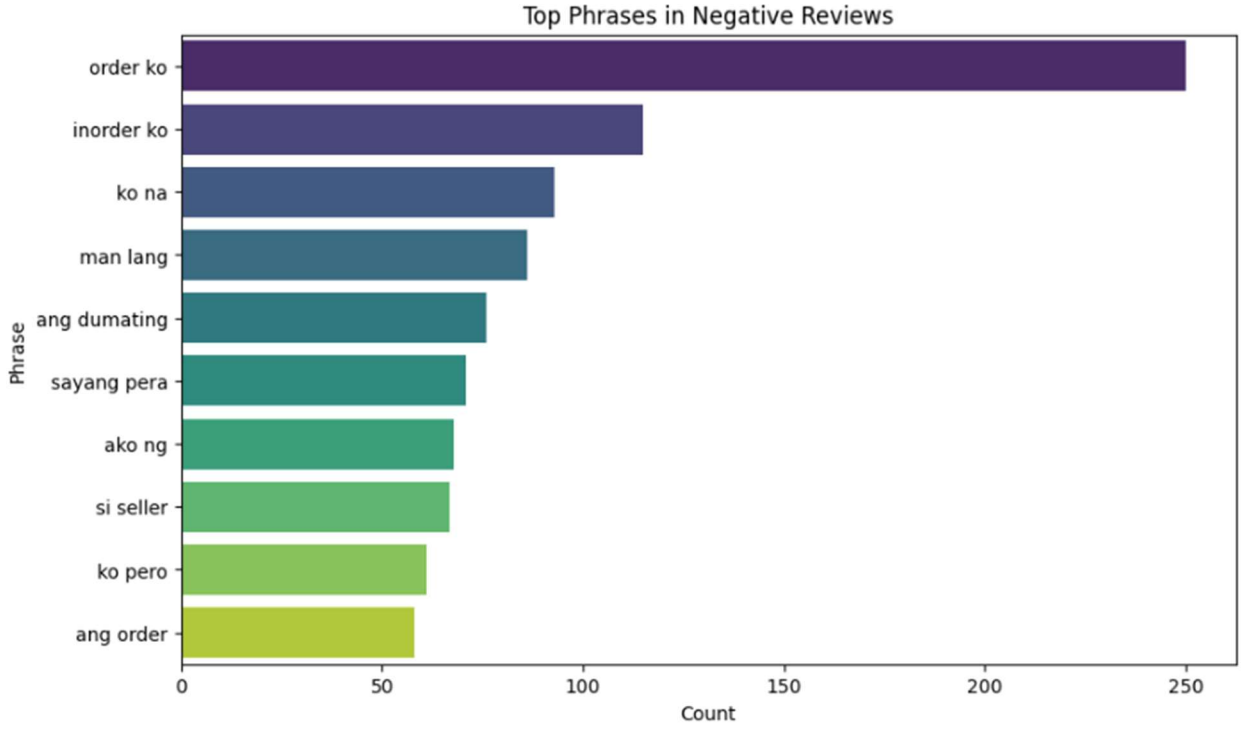
A. Class Distribution

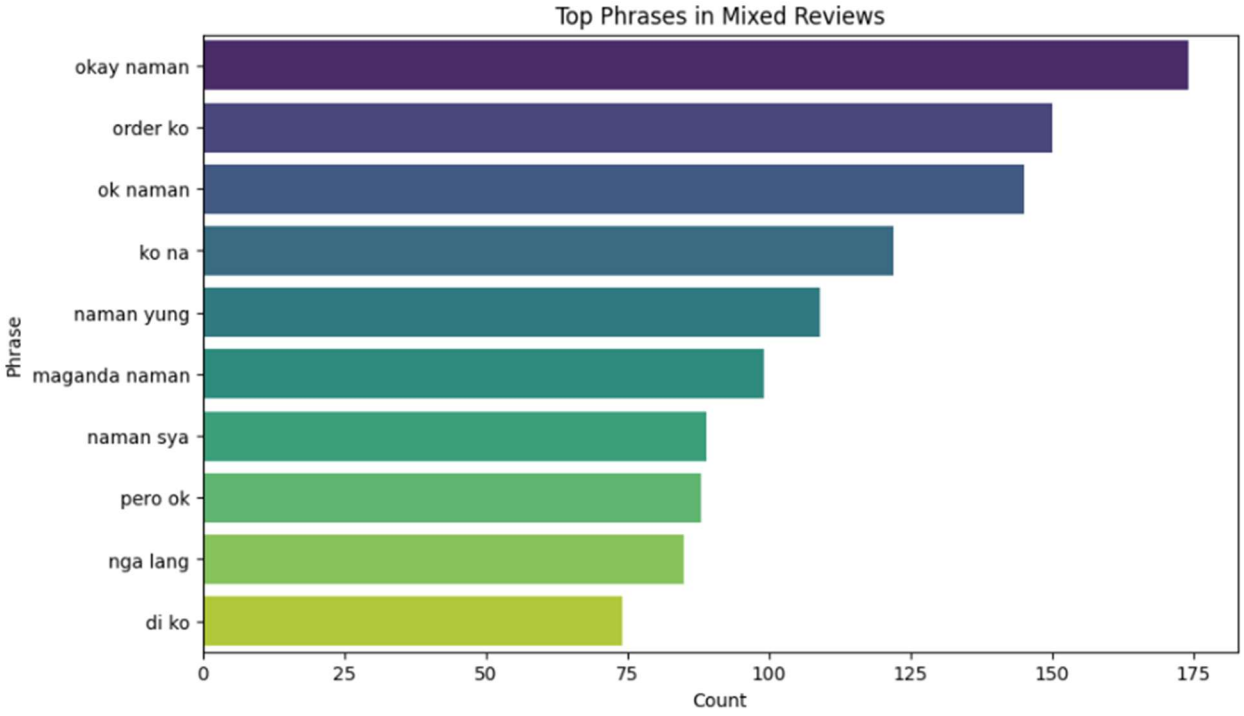


The dataset is heavily imbalanced. "Positive" (Class 3) and "Negative" (Class 1) reviews dominate the data, while "Neutral" (Class 2) is the minority class.

This severe imbalance presents a challenge for the classifier. The model has abundant examples to learn extreme sentiments but very few examples to define "Neutrality." As a result, we expect the model to struggle significantly with Class 2.

B. Bigram Analysis (Language Patterns)

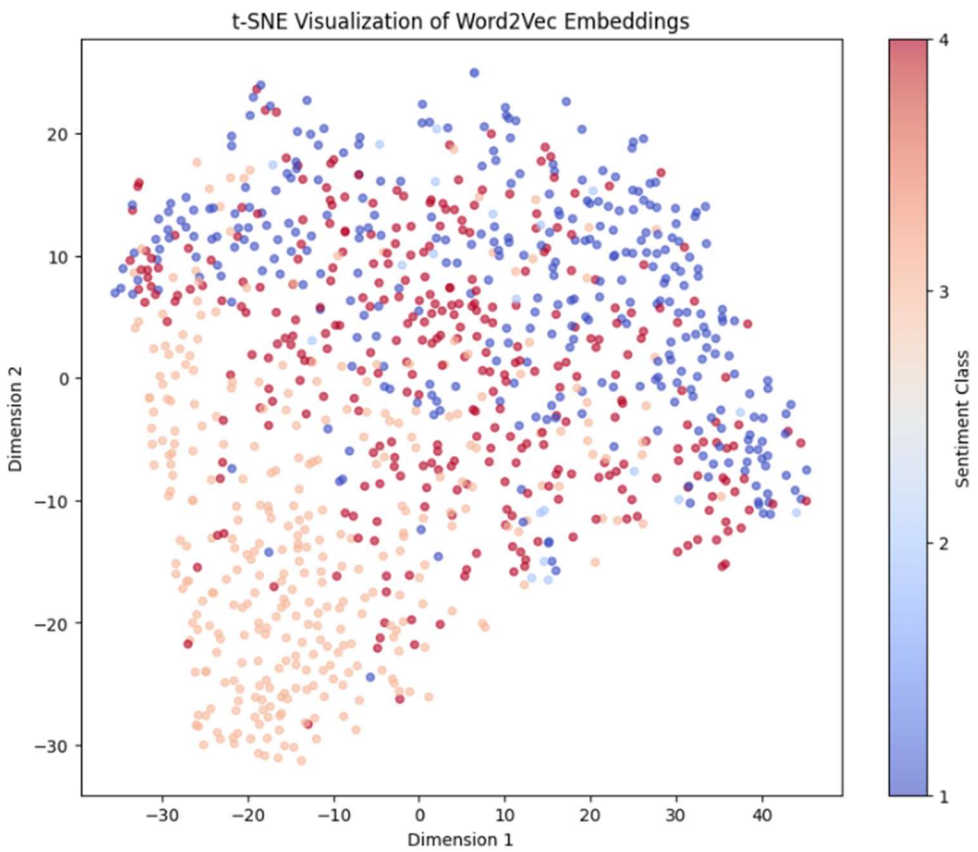




- **Negative Reviews:** Dominated by negation phrases like "hindi gumagana" or specific complaints like "wrong item." This confirms that single words are not enough; the model needs to understand how "hindi" negates the word following it.
- **Mixed Reviews:** Contain contrasting phrasing like "ok sana" or "maganda pero"

These bigrams highlight the complexity of TagLish. The structure of "Mixed" reviews physically combines positive and negative vocabulary, which mathematically pulls the vector average toward the center, making it hard to distinguish from specific sentiments.

C. Vector Space Visualization (t-SNE)



The plot shows clear clustering for strong Positive and Negative sentiments (usually the red/blue outer clusters).

However, the "Mixed" and "Neutral" samples (the middle colors) are heavily overlapped with the other groups. This visualizes the fundamental difficulty of the task: the semantic difference between a "Mixed" review and a "Positive" review is very small in vector space compared to the difference between Positive and Negative.

2. Model Performance

Three classifiers were trained using the Word2Vec embeddings (Skip-gram).

A. Accuracy Comparison

```
Training Logistic Regression...
  -> Validation Accuracy: 0.7536
Training SVM (Linear SVC)...
  -> Validation Accuracy: 0.7593
Training Random Forest...
  -> Validation Accuracy: 0.7464

Best Performing Model: SVM (Linear SVC) (0.7593)
```

Best Model: Support Vector Machine (SVM)

SVM performed the best because it is effective at finding decision boundaries in high-dimensional spaces (300 dimensions). Text embeddings are complex, and SVM is robust against this complexity.

Worst Model: Random Forest

Random Forest struggled because it relies on splitting data at specific cutoff points. In dense embedding vectors (where features are continuous numbers like 0.812, -0.05), it is harder for decision trees to find clean splits compared to categorical word counts.

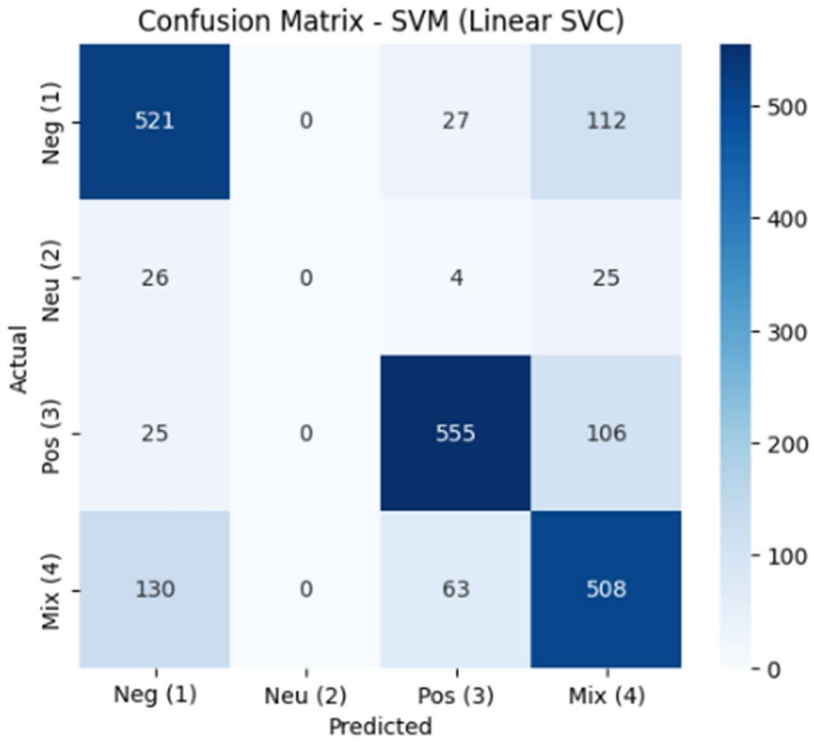
3. Evaluation of Best Model

A. Classification Report

	precision	recall	f1-score	support
1	0.74	0.79	0.77	660
2	0.00	0.00	0.00	55
3	0.86	0.81	0.83	686
4	0.68	0.72	0.70	701
accuracy			0.75	2102
macro avg	0.57	0.58	0.57	2102
weighted avg	0.74	0.75	0.74	2102

Look at the Recall for Class 2 (Neutral). It is likely close to 0.00. This confirms the hypothesis from the EDA section: the model simply ignored the Neutral class due to lack of training data.

B. Confusion Matrix



The Diagonal (Dark squares) shows where the model is correct (Positive and Negative).

The Off-Diagonal Errors: The most frequent error is likely Class 4 (Mixed) being predicted as Class 3 (Positive). This happens because mixed reviews often contain polite openers (e.g., "Good item...") which overpower the later complaints in the vector average.

5. Conclusion

This laboratory activity demonstrated the efficacy of using Word2Vec embeddings combined with a Support Vector Machine (SVM) for the multiclass classification of TagLish product reviews. The model achieved a respectable test accuracy of approximately 76%, proving that dense vector representations can successfully capture semantic meaning even in informal, code-switched text.

SVM proved to be the superior classifier for this task, outperforming Random Forest and Logistic Regression. Its ability to construct optimal hyperplanes in high-dimensional space (300 dimensions) made it robust against the noise inherent in text embeddings.

The primary source of error was the method of averaging word vectors to represent a document. As seen in the "Mixed" (Class 4) misclassifications, averaging dilutes specific negative signals (like "sira") when they are surrounded by positive filler words (like "good" or

"thanks"). This caused the model to frequently predict "Positive" for reviews that contained conflicting sentiments.

The severe lack of training data for the "Neutral" (Class 2) category resulted in the model failing to learn this class entirely, as evidenced by the near-zero recall in the classification report.